DESCENT OF THE FLAT-AT-PRIMES-OVER-AN-IDEAL CONDITION ALONG A SURJECTIVE MAP ON THE CLOSED SETS

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Allowed standard prerequisites (editorial).

- Flatness at a prime descends along a flat local base change. Algebra, Lemma \ref{algebra-lemma-base-change-flat-up-down}, cited at proof lines 4135-4137. Carries flatness of the localised base changed module back to the localisation of the module over the base.
- Surjectivity of the induced map on the closed sets defined by the ideals. assumption (1) of the lemma, used at proof lines 4125-4127. Produces the prime of the base change lying over the given prime.

Source reference. Original lemma 05LL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7240–7240. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 052W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6584–6584. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00MQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 832–832. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $R \to S$ be a ring map. Let $I \subset R$ be an ideal. Let M be an S-module. Let $R \to R'$ be a ring map and $IR' \subset I' \subset R'$ an ideal such that*

(1) *the map $V(I') \to V(I)$ induced by $\mathrm{Spec}(R') \to \mathrm{Spec}(R)$ is surjective, and*
(2) *$R'_{\mathfrak{p}'}$ is flat over R for all primes $\mathfrak{p}' \in V(I')$.*

If (052W) holds for $(R' \to S \otimes_R R', I', M \otimes_R R')$, then (052W) holds for $(R \to S, I, M)$.

Proof. Assume (052W) holds for $(R' \to S \otimes_R R', IR', M \otimes_R R')$. Pick a prime $IS \subset \mathfrak{q} \subset S$. Let $I \subset \mathfrak{p} \subset R$ be the corresponding prime of R. By assumption there exists a prime $\mathfrak{p}' \in V(I')$ of R' lying over $\mathfrak{p}$ and $R_{\mathfrak{p}} \to R'_{\mathfrak{p}'}$ is flat. Choose a prime $\overline{\mathfrak{q}}' \subset \kappa(\mathfrak{q}) \otimes_{\kappa(\mathfrak{p})} \kappa(\mathfrak{p}')$ which corresponds to a prime $\mathfrak{q}' \subset S \otimes_R R'$ which lies over $\mathfrak{q}$ and over $\mathfrak{p}'$. Note that $(S \otimes_R R')_{\mathfrak{q}'}$ is a localization of $S_{\mathfrak{q}} \otimes_{R_{\mathfrak{p}}} R'_{\mathfrak{p}'}$. By assumption the module $(M \otimes_R R')_{\mathfrak{q}'}$ is flat over $R'_{\mathfrak{p}'}$. Hence Algebra, Lemma 00MQ implies that $M_{\mathfrak{q}}$ is flat over $R_{\mathfrak{p}}$ which is what we wanted to prove. □